

EXPERIMENT – 25

Implementation of RSA Encryption and Demonstration of Common Factor Attack

CODE:

```
#include <stdio.h>
```

```
int gcd(int a, int b)
```

```
{
```

```
    while (b != 0)
```

```
    {
```

```
        int t = b;
```

```
        b = a % b;
```

```
        a = t;
```

```
    }
```

```
    return a;
```

```
}
```

```
int main()
```

```
{
```

```
    int n, e, message, factor;
```

```
    printf("Enter n and e: ");
```

```
    scanf("%d%d", &n, &e);
```

```
    printf("Enter known plaintext block: ");
```

```
    scanf("%d", &message);
```

```
    factor = gcd(message, n);
```

```

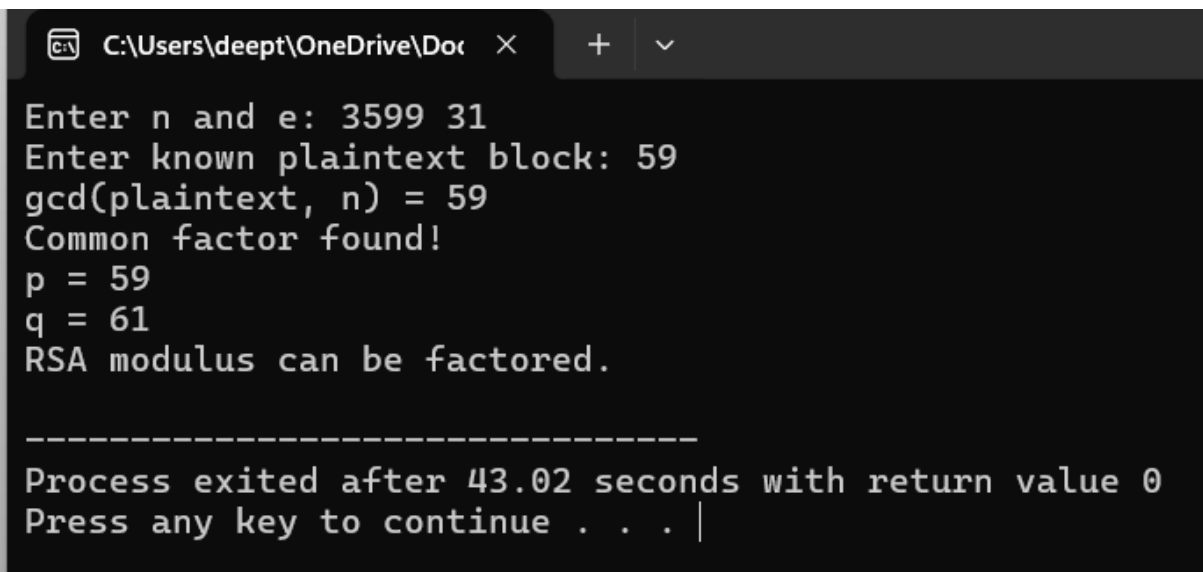
printf("gcd(plaintext, n) = %d\n", factor);

if (factor > 1 && factor < n)
{
    printf("Common factor found!\n");
    printf("p = %d\n", factor);
    printf("q = %d\n", n / factor);
    printf("RSA modulus can be factored.\n");
}
else
{
    printf("No useful common factor found.\n");
}

return 0;
}

```

OUTPUT:



```

C:\Users\deept\OneDrive\Doc > Enter n and e: 3599 31
Enter known plaintext block: 59
gcd(plaintext, n) = 59
Common factor found!
p = 59
q = 61
RSA modulus can be factored.

-----
Process exited after 43.02 seconds with return value 0
Press any key to continue . . . |

```